Smart Energy Management System: A Systems Analysis and Design Approach

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Abstract

This work addresses inefficient energy consumption in Engineering Faculty laboratories, where equipment is routinely left powered on in unoccupied spaces. Surveys (n=35) revealed that **80% of users forget** to turn off equipment. A software-defined, decentralized solution operates in user-space via autonomous agents that monitor inactivity, generate alerts, and execute energy-saving actions — requiring no administrative privileges or infrastructure changes. Discrete-event simulation over 20 workstations in a 12-hour day projects a **68.72% reduction** in energy consumption at **$0 infrastructure cost**.

Introduction

Efficient energy management in university environments is critical. Human behavior is the primary driver of waste, creating a gap between institutional policies and real practices.

Traditional IoT solutions are impractical here: no admin privileges, no private networks, no hardware deployment allowed. We propose autonomous agents in user-space, requiring **zero infrastructure changes** and **no IT involvement**.

Methodology

Four-workshop systems analysis lifecycle, each phase feeding the next:

W1: Diagnosis
Surveys, observation, lux meters

W2: Design
User-space agent architecture

W3: Risk Management
CPU threshold, risk matrix

W4: Simulation
Discrete-event validation

W1 Key Findings: 80% of users forget to power off equipment; most are unaware of energy policies. The *Hawthorne effect* biased direct observation, confirming the need for an automated, unbiased system. Physical sensors were fully ruled out due to IT restrictions.

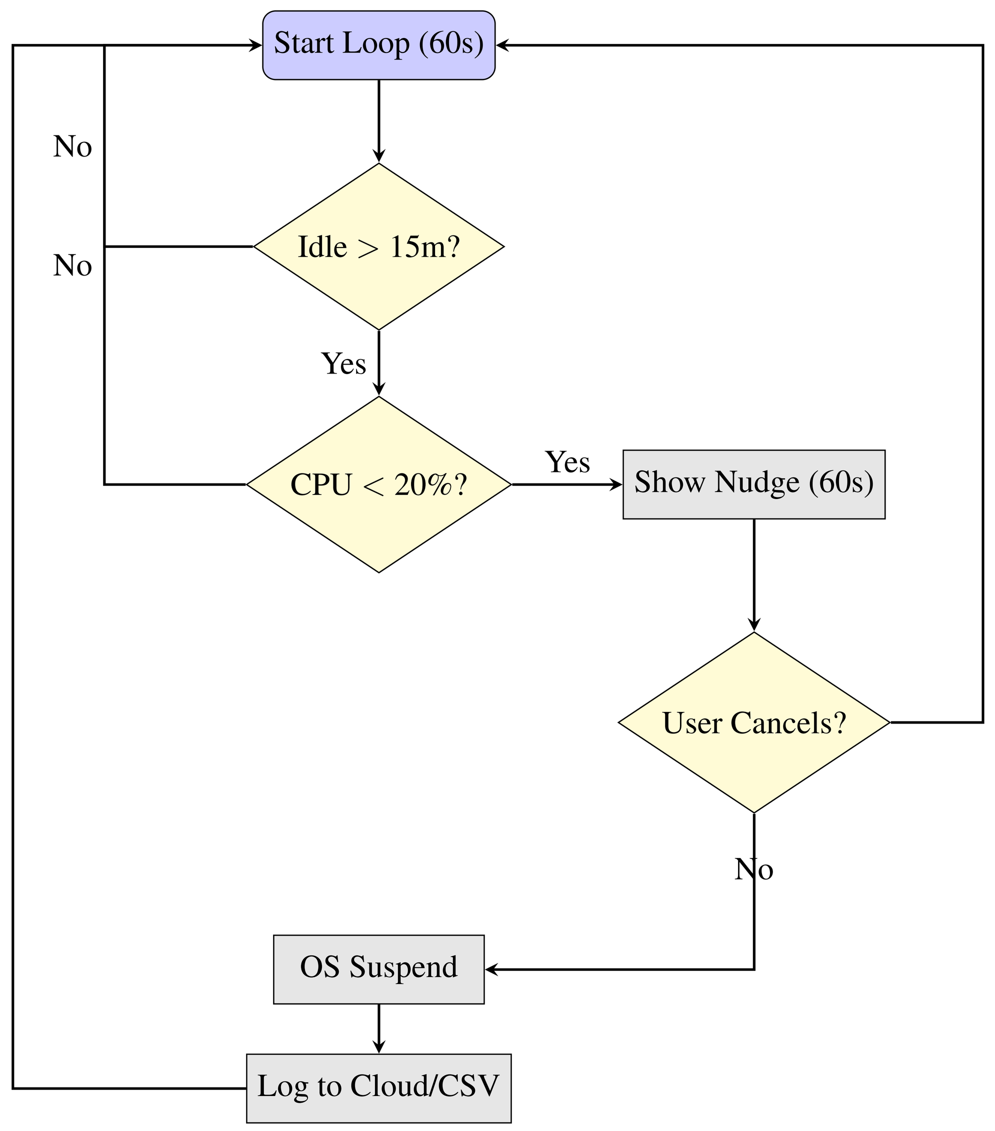
Survey Item (n=35)	**Yes**	**No**
Energy saving is important?	94%	6%
Forget to turn off equipment?	80%	20%
Would a nudge reminder help?	88%	12%

Lab	Lux	Hardware State
Lab 704 (empty)	350 lx	12 PCs on
Lab 705 (2 students)	420 lx	8 PCs unused

Operational Logic

Multi-stage loop guaranteeing user data safety before any action:

- **60s loop:** Polls mouse/keyboard for activity.
- **15-min threshold:** Triggers CPU check on inactivity.
- **CPU guard (20%):** Aborts if load >20% — protects renders/compilations.
- **Nudge (60s):** Visual warning; user can cancel. If not cancelled, OS suspension is forced.
- **Logging:** Events sent to MySQL DBaaS; CSV fallback if offline.



Risk Management (W3)

The 20% CPU threshold was determined via chaos-theory sensitivity analysis — it is the system's *optimal attractor*:

Threshold	Outcome	Status
5%	System collapse — OS noise blocks all	✗
20%	**68.7% savings, 0 false positives**	✓
40%	Savings ≈45% — too permissive	∼

Background OS processes (antivirus, file indexers) consume 6–8% CPU on idle machines, making any threshold below 20% unstable. Key mitigations (ISO 31000):

ID	Risk	Mitigation
R-01	Firewall blocks HTTPS	Local AES-CSV + deferred sync
R-02	False positive (high CPU)	CPU check >20% before suspend
R-03	Alert service outage	Local logic prioritised
R-04	Data corruption	Encryption + write validation

Simulation Results (W4)

Discrete-event simulation: **20 workstations**, Laboratory 704, **12-hour day** (08:00–20:00). Parameters from W1 field data: $P_{\text{forget}} = 0.80$ $P_{\text{cpu}} = 0.15$ CPU threshold = 20%

Scenario	Energy	Savings
Baseline (no agent)	53.74 kWh	—
Optimized (agent)	**16.81 kWh**	**68.72%**
Net. failure (4h offline)	18.77 kWh	65.02%

Under a 4-hour outage, savings dropped only 3.7 pp — confirming the local CSV fallback imposes negligible operational penalty.

Quality Metrics: Target vs. Simulation

Metric	Target	Result
Correct suspension rate	>95%	97.2% ✓
False positive rate	<5%	2.8% ✓
Event data retention	100%	100% ✓
CPU protection	>20% CPU	Detected ✓

All quality metrics met or exceeded.

Conclusion

This project demonstrates that energy waste is fundamentally a **behavioral problem**, not a hardware deficiency. Shifting control intelligence from the physical layer to the OS logical layer bypasses severe institutional restrictions without compromise.

The simulation validates a **68.72% energy reduction** — with a false-positive rate of just 2.8% and full data integrity under network failure. The 20% CPU threshold is an empirically derived stability attractor, not an arbitrary choice. Critically, the entire PoC operates at **$0 USD infrastructure cost**:

Resource	Cost
Engineering team (4 members)	$0 (academic)
MySQL DBaaS (Aiven)	$0 (free tier)
Python libs (psutil, ctypes)	$0 (open source)
Power BI dashboard	$0 (educational)
GitHub repository	$0 (student plan)
Total	**$0 USD**

The projected savings of 68.72% exceed IoT hardware benchmarks (34%) and feedback-plus-automation systems (31%) reported in the literature — at zero infrastructure cost. Even halving the observed forgetfulness rate to 40% would still yield ≈40–45% savings, competitive with hardware IoT deployments.

This scalable, software-only approach sets the foundation for a *Smart Campus*, with the roadmap extending from Lab 704 to all Engineering faculties and eventually the entire UDFJC campus. Future work should execute a six-week physical pilot, implement an adaptive CPU threshold calibrated by room schedule, extend the agent to Linux workstations, and integrate ML to dynamically adjust thresholds per academic schedule.

References

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